

Did Cutting the Universal Credit Taper Rate Change Who Works?

Testing whether the substitution effect vs. income effect dominated after the December 2021 Universal Credit taper reform, using UK administrative data.

R (fixest, tidyverse)

SQL

PostgreSQL

DiD

Two-Way Fixed Effects (TWFE)

Event-Study Design

Clustered Standard Errors

Causal Inference

Panel Data

Applied Microeconometrics

Labour Economics

Policy Evaluation

HMT Magenta Book

Government Administrative Data

Reproducible Research

Question

In December 2021, the UK government cut the Universal Credit taper rate (the rate at which benefit is withdrawn as claimants earn more), from 63% to 55%. We can predict two competing responses in economic theory. This project tests which response actually dominated, using local-authority-level administrative data:

- Substitution effect: work now pays relatively more, so claimants have more reason to work extra hours.
- Income effect: claimants keep more of what they already earn, so some can work less for the same income.

Data & Methods

The full set of code scripts, methodology and results can be viewed directly on GitHub:

<https://github.com/sophia-adams1/uc-taper-rate-reform>

Datasets

- **DWP Stat-Xplore** - Monthly UC claimant counts by local authority, by employment status
- **NOMIS (Annual Population Survey)** - Local authority employment rate, 2021 & 2022
- **ONS ASHE** - Local authority median hourly pay and weekly hours, 2021 (pre-reform)
- **ONS Geography Lookup** - Output Area → Local Authority → Travel-to-Work Area crosswalk

Methodology

A two-way fixed effects difference-in-differences design was used with an event-study extension to estimate the causal effect of the December 2021 Universal Credit taper rate cut on claimant employment. Four public data sources (see above) were cleaned and joined via a SQL pipeline into a single local authority by month panel, with local authorities classified by pre-reform (2021) median pay to construct a treatment group unaffected by post-reform outcomes. Two-way fixed effects models were estimated in R using fixest, with standard errors clustered at Travel-to-Work Area to account for shared local labour markets, and an event-study specification was run to formally test the parallel trends assumption and trace the effect month by month rather than as a single average. During development, a duplicate-row issue in the geography join was identified and corrected after it was found to be inflating the sample and artificially producing a false-positive significant result; the corrected pipeline shows a significant rise in both in-employment and not-in-employment claimant counts in lower-pay areas, but no significant change in the employment share, a result reported as the genuine finding rather than the more dramatic uncorrected one.

Findings

Claimant counts rose significantly in lower-pay areas after the reform for both in-employment and not-in-employment claimants ($p=0.03$, $p=0.04$). The employment share itself, however, did not move significantly ($p=0.48$). Together, this points to broader caseload growth in lower-pay areas, rather than a genuine shift in who was actually working. This is a more precise conclusion than either theory alone predicted, and the one the data actually supports.

Share of UC claimants in employment, by exposure group

Dashed line marks the December 2021 taper rate reform

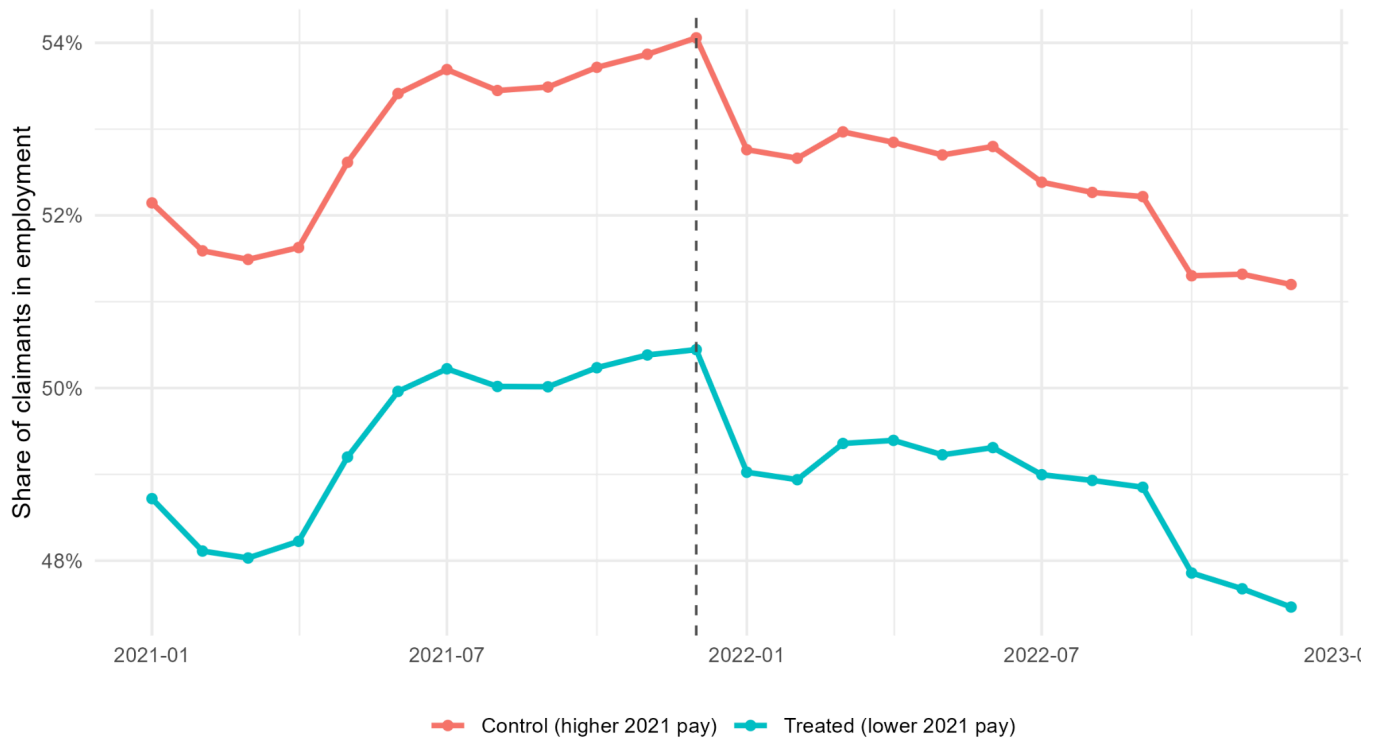


Fig. 1. Control and treated areas move together until the reform, then diverge and stay apart. This is the pattern a genuine treatment effect should produce.

Event-study: effect of taper cut on share of claimants in employment

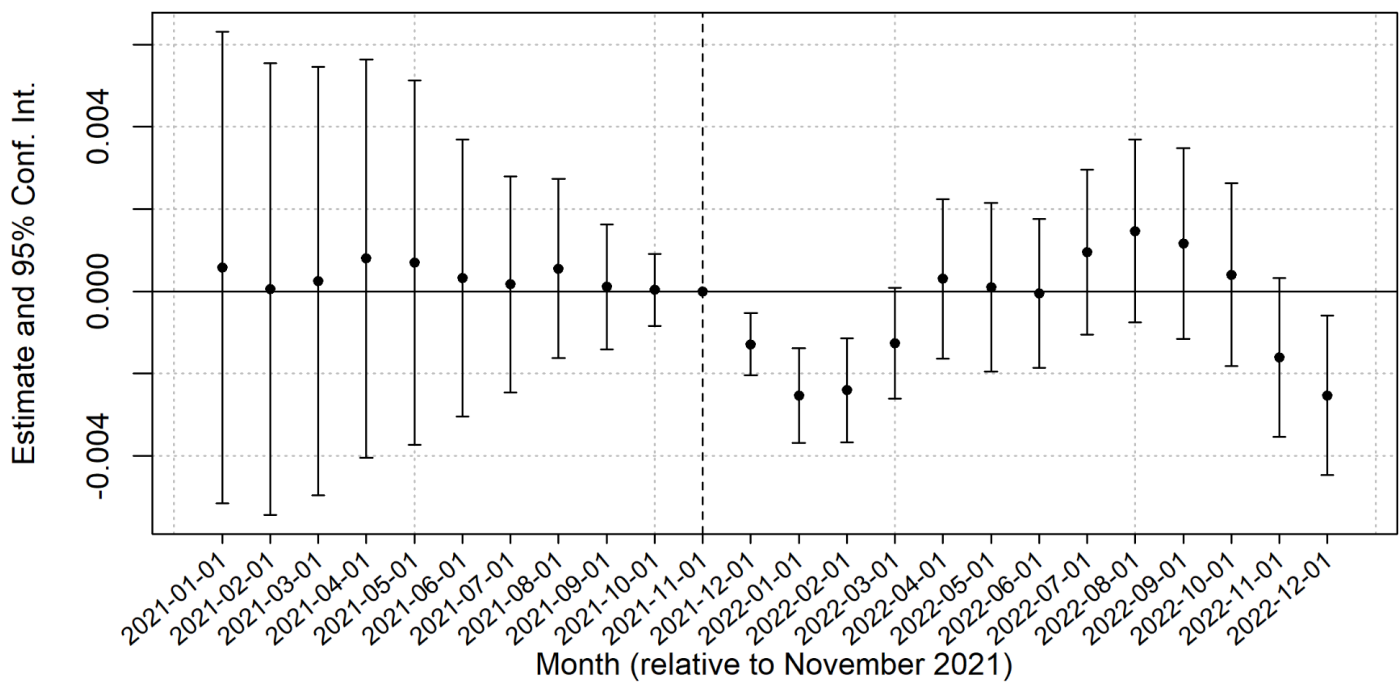


Fig. 2. Month-by-month effect estimates with 95% confidence intervals. Pre-reform estimates cross zero, which supports the parallel trends assumption underlying the design.